

Lab 7: Energy, Photosynthesis, & Cellular Respiration

BIOL-1

Name: _____ **Date:** _____

Overview

Energy is the currency of life. It flows through ecosystems from the sun to producers, then to consumers, and eventually dissipates as heat. In contrast, matter (like carbon, hydrogen, and oxygen) constantly cycles between the environment and living organisms. In this comprehensive lab, you will model the complementary processes of photosynthesis and cellular respiration. Through detailed structural drawing and step-by-step pathway tracking, you will build a robust understanding of how cells capture, transfer, and consume energy.

Learning Objectives

By the end of this lab, you will be able to:

1. Relate the balanced chemical equations for photosynthesis and cellular respiration to the cycling of carbon.
2. Trace the path of energy from sunlight to stored chemical energy (glucose), and finally to usable cellular energy (ATP).
3. Diagram the microscopic structures of the chloroplast and mitochondrion and identify where specific metabolic reactions occur.
4. Model the Electron Transport Chain (ETC) and explain how chemiosmosis generates ATP.
5. Calculate the relative energy yields of aerobic respiration versus anaerobic fermentation under different physiological conditions.
6. Connect the concepts of autotrophy and heterotrophy to global ecological systems.

Materials

- Review Packet & Activity Worksheets (this document)
- Colored pencils (Green, Red, Blue, Yellow, Black)
- Reference textbook or lecture notes on Metabolism

Part 1: The Anatomy of Energy Organelles

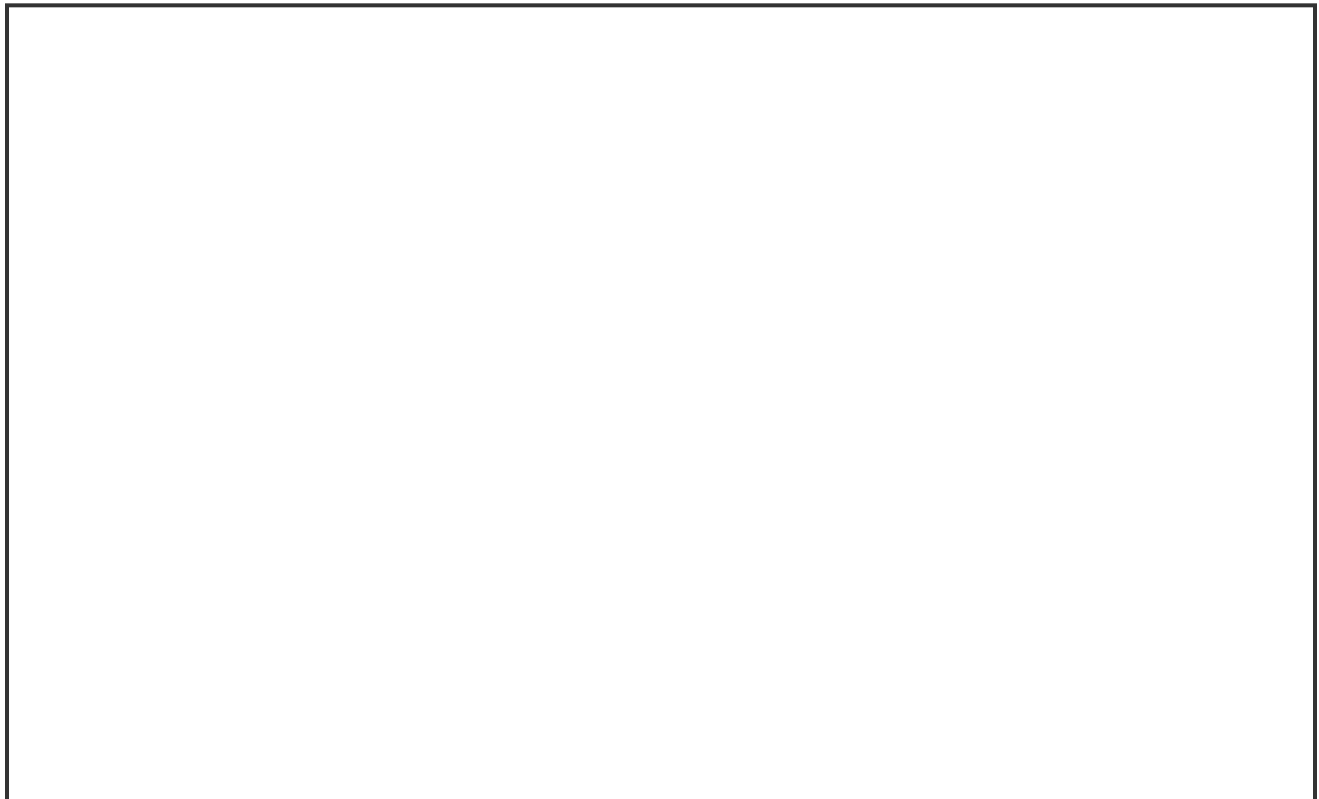
Form dictates function in biology. The intricate folding of biological membranes in chloroplasts and mitochondria maximizes the surface area available for the enzymatic machinery of energy conversion.

Box A: The Chloroplast

Draw a detailed chloroplast. Label the **Outer Membrane**, **Inner Membrane**, **Stroma**, **Thylakoid**, and **Granum**. Use a green pencil to shade the areas where chlorophyll is concentrated.

Box B: The Mitochondrion

Draw a detailed mitochondrion. Label the **Outer Membrane**, **Inner Membrane**, **Cristae**, and **Matrix**. Use a red pencil to trace the highly folded inner membrane to emphasize increased surface area.



1. How does the highly folded structure of the cristae in the mitochondrion directly relate to its primary function?

2. Which part of the chloroplast corresponds anatomically to the mitochondrial matrix? (i.e., the innermost fluid-filled space where a major metabolic cycle occurs)

Part 2: Photosynthesis — Capturing Solar Energy

Photosynthesis converts solar energy into stable chemical energy stored in the bonds of glucose. It occurs in two main stages: the Light-Dependent Reactions and the Calvin Cycle (Light-Independent Reactions).

2A. The Light Reactions (The "Photo" Part)

Draw a section of the thylakoid membrane. Include and label:

- Photosystem II and Photosystem I
- The Electron Transport Chain (ETC) proteins
- ATP Synthase
- The path of electrons (e^-) from H_2O to NADPH
- The pumping of protons (H^+) into the thylakoid space

3. Water (H₂O) is split in Photosystem II. What three things are produced when water is split, and where does each one go?

2B. The Calvin Cycle (The "Synthesis" Part)

The ATP and NADPH generated above are sent to the stroma to power the Calvin Cycle, which "fixes" inorganic CO₂ gas into solid sugar.

4. It takes 6 molecules of CO₂ to build one molecule of glucose (C₆H₁₂O₆). If a plant leaf has its stomata (pores) closed during a hot, dry day to prevent water loss, what happens to the rate of the Calvin Cycle, and why?

Part 3: Cellular Respiration — Releasing Chemical Energy

Cellular respiration breaks down glucose to harvest its stored energy, producing ATP that the cell can use for immediate work. It happens in three main stages: Glycolysis, the Krebs Cycle, and Oxidative Phosphorylation.

3A. Tracking the Carbon

Glycolysis breaks the 6-carbon glucose into two 3-carbon pyruvate molecules. Pyruvate oxidation and the Krebs cycle then completely dismantle the pyruvates into CO₂.

5. Complete the table: Where does each stage occur, and is oxygen required?

Stage	Location in Cell	Requires O ₂ ? (Yes/No)
Glycolysis		
Pyruvate Oxidation		
Krebs Cycle		
Electron Transport Chain		

6. By the end of the Krebs Cycle, what has happened to all six carbon atoms from the original glucose molecule?

3B. The Grand Finale: Oxidative Phosphorylation

Draw a section of the mitochondrial Inner Membrane (Cristae). Include and label:

- Protein Complexes I, II, III, and IV
- ATP Synthase
- NADH delivering electrons
- Oxygen (O₂) catching electrons at the end to form Water (H₂O)

- Protons (H^+) being pumped into the intermembrane space

7. Compare your drawing here to the one in Part 2A. Briefly explain the concept of *chemiosmosis* as a universal mechanism for making ATP in both organelles.

8. Cyanide is a deadly poison that blocks Protein Complex IV in the ETC. Explain precisely why cyanide rapidly kills a cell.

Part 4: Fermentation Case Studies

When oxygen is absent, the ETC shuts down, halting the Krebs cycle. Cells must rely on Glycolysis alone, followed by fermentation to regenerate NAD^+ .

Case Study A: The Sprinter

A track athlete runs a 400-meter dash. Around the 200-meter mark, her muscle cells use up available oxygen faster than her cardiovascular system can deliver it.

- **Pathway:** Her cells switch to Lactic Acid Fermentation.
- **Energy Yield:** Only 2 ATP per glucose molecule.

Case Study B: The Brewer's Yeast

Yeast cells are sealed in a vat of sugary grape juice with no oxygen.

- **Pathway:** The yeast cells perform Alcoholic Fermentation.
- **Energy Yield:** 2 ATP per glucose, plus Ethanol and CO₂ (bubbles).

9. A single glucose molecule can yield ~32 ATP via aerobic respiration. If the sprinter's muscle cell urgently needs 320 ATP, how many glucose molecules must it burn if it is strictly using lactic acid fermentation?

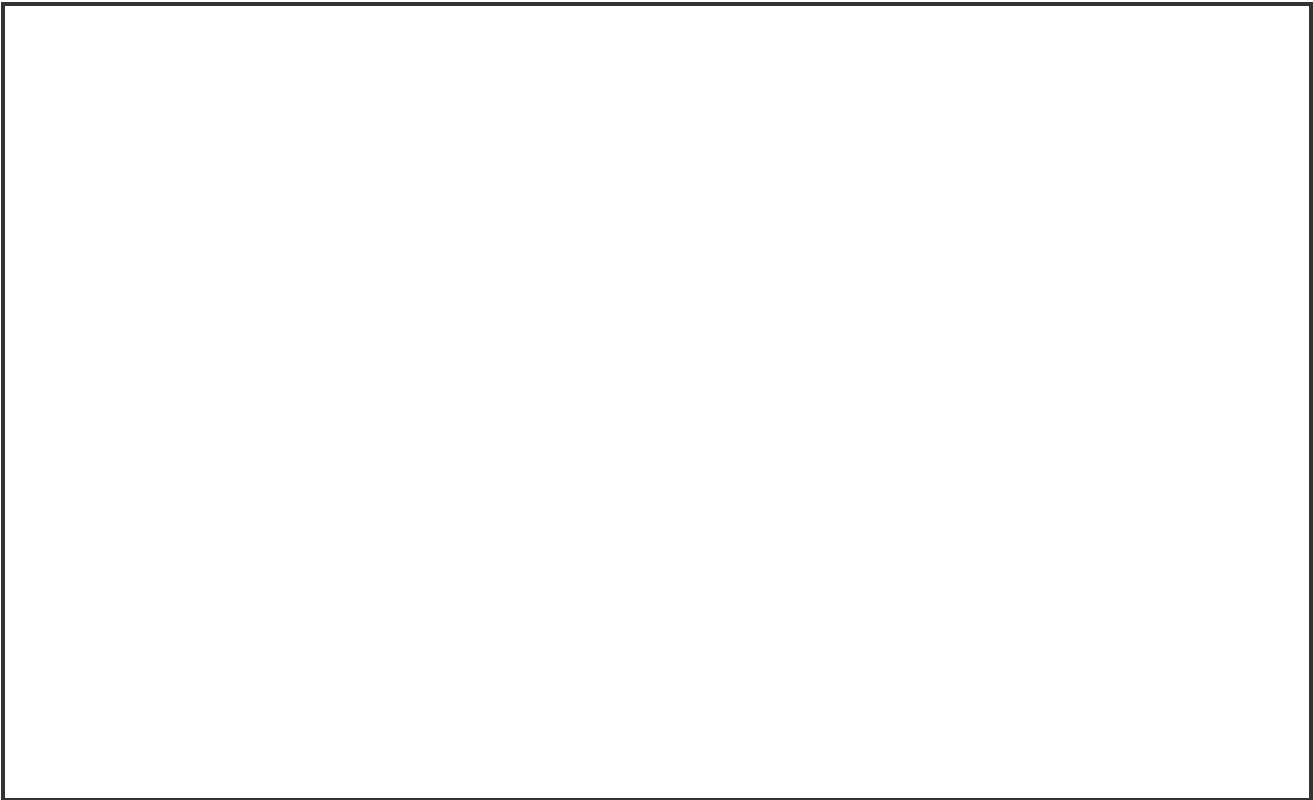
10. Why do the athlete's muscles undergo fermentation rather than simply stopping all metabolism until oxygen returns?

Conclusion: The Grand Cycle

Complete the master diagram below to show the macroscopic relationship between the two processes.

Draw a large cycle connecting a plant (autotroph) and an animal (heterotroph).

- Draw the sun providing energy to the plant.
- Draw arrows showing the flow of the required inputs and the outputs for both organisms.
- Label the arrows with: **Glucose, O₂, CO₂, H₂O, Heat, and ATP.**



11. "Energy flows, but matter cycles." Defend this statement using the diagram you just drew.

